

STAT lecture 2

Discrete random variables and probability distributions

Disclaimer:

Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples

Discrete random variable

Stochastic experiment

A random variable, X , defines a real number for each of the outcomes in sample space, S , i.e. $X: S \rightarrow I \subseteq \mathbb{R}$

e.g. outcome a_1 : $X = x_1$

Discrete random variable: the outcomes are countable (finite or infinite)

Probability mass function, pmf

Defines the probability for all possible outcomes in S

$$f_X(x_i) \stackrel{\text{def}}{=} P(X = x_i)$$

$$0 \leq f_X(x_i) \leq 1$$

$$\sum_S f_X(x_i) = 1$$

Cumulative ("added from left") probability mass function, cmf

$$F_X(x_i) \stackrel{\text{def}}{=} P(X \leq x_i) = \sum_{x_j \leq x_i} P(X = x_j) = \sum_{x_j \leq x_i} f_X(x_j)$$

$$0 \leq F_X(x) \leq 1$$

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

$$P(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

Mean value of X

$$\mu_X = E[X] \stackrel{\text{def}}{=} \sum_{x \in S} x f_X(x)$$

Variance of X

$$\sigma_X^2 = V[X] \stackrel{\text{def}}{=} E[(X - \mu_X)^2] = E[X^2] - \mu_X^2 \geq 0$$

Discrete uniform distribution

$$X \sim UD(a, b), \quad a, b \in \mathbb{Z}$$

$$f_X(x) = \frac{1}{b-a+1}, \quad a \leq x \leq b$$

$$\mu_X = \frac{a+b}{2}$$

$$\sigma_X^2 = \frac{(b-a+1)^2 - 1}{12}$$

Binomial distribution

$$X \sim Bin(n, p), \quad n \in \mathbb{N}, \quad 0 \leq p \leq 1$$

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$\mu_X = np$$

$$\sigma_X^2 = np(1-p)$$

Geometric distribution

$$\begin{aligned} X &\sim \text{Geo}(p), \quad 0 \leq p \leq 1 \\ f_X(x) &= (1-p)^{x-1} p, \quad x = 1, 2, \dots \\ \mu_X &= \frac{1}{p} \\ \sigma_X^2 &= \frac{1-p}{p^2} \end{aligned}$$

Poisson distribution

$$\begin{aligned} X &\sim \text{Poisson}(\lambda), \quad \lambda \geq 0 \\ f_X(x) &= \frac{(\lambda t)^x}{x!} e^{-\lambda t}, \quad x = 0, 1, 2, \dots \\ \mu_X &= \lambda t \\ \sigma_X^2 &= \lambda t \end{aligned}$$